# Unsupervised Deep Learning for Ecological Site Groups in MLRA 42B

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## Problem Statement

Across the western United States, land managers, ranchers, and conservation agencies rely on a system of land classification called Ecological Sites. An Ecological Site is a patch of landscape with a distinct combination of soil, climate, and topography that determines what plants can grow there and how the land responds to disturbance. It describes the whole soil-climate-vegetation system and how it can shift between different states over time (Karl and Herrick, 2010; Bestelmeyer et al., 2003). Ecological site maps are built largely by hand: soil scientists and ecologists visit sites, take measurements, and use expert judgment to draw boundaries and assign classifications. This is accurate but slow and expensive, and in practice ecological site maps are produced by linking ecological sites to soil survey (SSURGO) map units rather than being mapped independently.

The ecological site concept has become cumbersome; many sites with different names have properties that are functionally quite similar. The USDA would like to transition from ecological sites to ecological site groups, effectively absorbing ecological sites with similar properties into broader classification categories. This project asks whether an unsupervised deep learning model can divide MLRA 42B (Southern Rio Grande Rift) into ecologically meaningful groups using only environmental covariates. We then ask where those groups agree and disagree with mapped ecological sites. Disagreement is the main interest: it may give insight into existing map-unit uncertainty, ecological site concepts that don't fit the landscape, or gaps in our covariates or model.

## Motivation & Objectives

Deep learning fits this problem for three reasons. The inputs are high-dimensional and multisource. Ecologically important patterns are spatial (landform position, drainage, soil properties) and are lost when pixels are treated independently (Padarian et al., 2019; Wadoux et al., 2019). The goal is to discover groups without labels, which suits representation learning. Prior digital ecological-site mapping has been supervised and trained on existing ESG classes (Maynard et al., 2019). Our unsupervised approach does not assume current ecological site concepts capture all the heterogeneity of landscape patterns. The task type is unsupervised clustering.

**Objectives:**

1. Use a convolutional autoencoder to condense the stacked landscape data and capture its key environmental and spatial patterns.
2. Define the number of clusters and produce a map of ecological groups.
3. Describe each group by the range of its covariates.
4. Measure agreement with SSURGO-mapped ecological sites and characterize where and why they disagree.
5. Depending on time, test whether groups differ in vegetation response to drought using Rangeland Analysis Platform (RAP) cover time series.

## Dataset Description

The study extent is MLRA 42B plus a 5 km buffer, which gives image patches at the MLRA edge complete spatial context. All data are public except the STEDIS terrain and spectral derivatives, which are derived from public data (USGS 3DEP elevation and Landsat imagery) but are not yet published. We assembled 23 candidate raster layers from four sources. After screening for redundancy (see Preprocessing), **19 layers form the final covariate stack** (Table 1). Model inputs and validation data are kept separate.

**Table 1.** Covariate sources, candidate layers, and layers retained in the final stack.

| Group | Source | Native resolution | Candidate layers | In final stack |
|---|---|---|---|---|
| Terrain | STEDIS terrain derivatives (from USGS 3DEP DEM) | 30 m | 8 | 7 |
| Spectral | STEDIS spectral derivatives (from Landsat) | 30 m | 3 | 3 |
| Climate | PRISM 1991–2020 30-year normals | ~800 m | 5 | 3 |
| Soil | SOLUS (Nauman et al., 2024) | 100 m | 7 | 6 |
| **Total** | | | **23** | **19** |

### *Layers in the covariate stack*

Table 2 describes each layer in the final stack and why it is relevant to ecological sites in MLRA 42B, a region of desert basins, alluvial fan piedmonts, and isolated mountain ranges in southern New Mexico and west Texas, where carbonate accumulation, soil depth, texture, and water redistribution strongly shape plant communities (Gile et al., 1981). For STEDIS layers, the name suffix gives the processing parameter: "_32" layers were computed over a 32-pixel (~960 m) neighborhood; "_10" and "_5" are parameters of the wetness and channel-distance algorithms. SOLUS layer names give the property, the depth in cm, and the estimate type ("_p" = prediction).

**Table 2.** The 19 layers in the final covariate stack.

| Layer (file name) | What it is | Why it is used |
|---|---|---|
| **Terrain (STEDIS, 30 m)** | | |
| sl_32 | Slope: steepness of the land surface. | Controls runoff versus infiltration, erosion, and soil depth; separates mountain and hill slopes from piedmonts and basin floors. Slope class is a standard ecological site criterion. |
| aspct_32 | Aspect: compass direction the slope faces (0–360°). | North- versus south-facing slopes differ in solar exposure, soil temperature, and moisture, which matters on mountain and hill slopes. In this version aspect entered as raw degrees (see Known issues). |
| perctelev_32 | Percentile elevation: rank of a cell's elevation within its surrounding neighborhood (0–100). | Identifies relative landscape position (summits and shoulders versus footslopes and basin bottoms) independent of absolute elevation. |

| Layer (file name) | What it is | Why it is used |
|---|---|---|
| relmeanelev_32 | Relative mean elevation: difference between a cell's elevation and the mean elevation of its neighborhood. | Positive values mark convex, shedding positions; negative values mark concave, collecting positions. Distinguishes run-off from run-on settings. |
| tpi_32 | Topographic position index: a standardized measure of whether a cell sits above or below its surroundings (Weiss, 2001). | A widely used landform-position metric separating ridges, slopes, and valleys. Related to relative mean elevation (r = 0.73 in screening) but retained by the redundancy screen. |
| swi_10 | SAGA wetness index: potential soil moisture from upslope contributing area and slope, with broad valley floors treated as uniformly wet (Böhner et al., 2002; Conrad et al., 2015). | Water redistribution is a primary control in drylands. Run-on positions such as draws, swales, and playa margins support distinct ecological sites (e.g., draw and bottomland sites). |
| vdcn_5 | Vertical distance to channel network: height of a cell above the nearest drainage line (Conrad et al., 2015). | Separates floodplains and low terraces near channels from fan remnants and uplands; relates to flooding, sediment deposition, and access to extra water. |
| **Spectral (STEDIS, from Landsat, 30 m)** | | |
| US_6Band_ChangeFree_mosaic_B5 | Band 5 of a 6-band Landsat "change-free" composite (a multi-year mosaic excluding pixels affected by land-cover change). In a 6-band blue-to-SWIR stack this is shortwave infrared 1 (~1.55–1.75 µm). | Sensitive to soil surface brightness, moisture, mineralogy, and dry plant litter. With sparse vegetation, the soil surface dominates the signal, helping separate gravelly, calcareous, and sandy surfaces. |
| NDVI_90_IQR | Interquartile range of NDVI (vegetation greenness) over the Landsat time series. | Measures how strongly greenness fluctuates; separates rain-responsive grass-dominated systems from more stable shrublands. |
| NDVI_Trend_post2000_linearFit | Linear trend in NDVI since 2000. | Captures long-term greening or browning, for example shrub encroachment or degradation. Reflects recent change and current state rather than site potential (see caveat below). |
| **Climate (PRISM 1991–2020 normals, ~800 m)** | | |
| prism_ppt | Mean annual precipitation (mm). | Water is the primary limit on plant production in drylands; precipitation separates arid basin sites from wetter mountain sites, and ecological sites are partly organized by precipitation zone. |
| prism_tmax | Mean daily maximum temperature (°C). | Drives evaporative demand and plant water stress, and separates hot basin floors from cooler uplands. Retained as the single temperature variable after mean and minimum temperature were removed as redundant. |
| prism_solclear | Clear-sky solar radiation. | Energy available for evaporation and plant growth. In 42B it varies by less than 1% (see Known issues). |

| Layer (file name) | What it is | Why it is used |
|---|---|---|
| **Soil properties (SOLUS, 100 m)** | | |
| anylithicdpt_cm_p | Predicted depth to any lithic (bedrock) contact (cm). | Soil depth limits rooting depth and water storage and separates shallow upland sites from deep soils. Most of 42B is near the maximum predicted depth (mean 191 cm, SD 13 cm), so the layer mainly identifies shallow soils. |
| caco3_30_cm_p | Calcium carbonate equivalent (%) at 30 cm. | Shallow calcic horizons are a defining feature of southern New Mexico basin soils. |
| caco3_150_cm_p | Calcium carbonate equivalent (%) at 150 cm. | Captures deeper carbonate accumulation, including petrocalcic horizons on older geomorphic surfaces. Together with the 30 cm layer it describes the carbonate depth profile. |
| claytotal_30_cm_p | Clay content (%) at 30 cm. | Subsoil texture controls water-holding capacity and the presence of argillic horizons, separating sandy, loamy, and clayey sites. |
| fragvol_0_cm_p | Rock fragment volume (%) at the surface. | Marks gravelly fan piedmonts and desert pavements; affects infiltration and erosion and defines gravelly ecological sites. |
| fragvol_30_cm_h | Rock fragment volume (%) at 30 cm, upper horizon estimate. | Subsurface gravel content affects water storage and rooting. |

**Validation data (not model inputs).** Ecological sites come from SSURGO soil map unit components linked to Ecological Site Descriptions (EDIT); disagreement between these maps and our groups is the main interest.

**Caveat on SOLUS.** SOLUS layers are model predictions, partly trained on field observations paired with soil survey component estimates (Nauman et al., 2024). They are therefore not fully independent of the SSURGO-linked ecological site map used for validation. We plan to run the model with and without the SOLUS layers to measure how much they drive agreement with mapped ecological sites.

**Caveat on spectral layers.** The three spectral layers describe current vegetation and surface condition, not site potential, so they may group areas by condition (state) rather than by ecological site.

## Preprocessing (Completed)

Preprocessing is implemented in R with the terra package as two scripts in the repository's src folder.

### Step 1: Clipping (01_clip_covariates.R)

- The MLRA 42B polygon was selected from the NRCS MLRA boundary file, dissolved into one polygon, projected to CONUS Albers (EPSG:5070), and buffered by 5 km.
- Each CONUS-wide raster was cropped and masked to the buffered area in its own projection and resolution, so no values were altered at this step. The script logs each layer's projection, resolution, size, and data type, skips finished layers when rerun, and records failures without stopping.

- The SOLUS download included auxiliary files (validation plot images and training-point density rasters) that are not soil maps. These failed because they lack a coordinate system and were excluded; only SOLUS prediction layers were used.
- The STEDIS terrain layers carry a projection without a standard EPSG code, so the final grid was copied directly from the slope layer rather than defined by an EPSG code.

### Step 2: Building the stack (02_build_covariate_stack.R)

- **Manifest:** a CSV listing all 23 candidate layers with their file paths and types. An early run mislabeled several integer-valued SOLUS layers as categorical, which would have split them into one-hot class channels; the manifest was corrected so that all 23 layers are continuous.
- **Common grid:** all layers were aligned to one 30 m grid matching the STEDIS terrain grid, so terrain layers were not resampled. Coarser layers (PRISM ~800 m, SOLUS 100 m) were resampled to 30 m with bilinear interpolation; they remain smooth and gain no real fine-scale detail.
- **Missing data:** any cell missing in one layer was set to missing in all layers, so every cell has either a complete set of values or none.
- **Statistics sample:** 40,000 random complete cells were used for redundancy screening and standardization.
- **Redundancy screen:** a two-tier variance inflation factor (VIF) procedure removed near-duplicate layers. VIF measures how well a layer can be predicted from all other layers combined (VIF = $1/(1 - R^2)$). Layers were removed one at a time, highest VIF first, until all remaining layers had VIF $\leq$ 10 (tier 1) and then VIF $\leq$ 5 (tier 2). Because no thematic groups were assigned, both tiers were applied to all layers. Four layers were removed (Table 3).
- **Standardization:** each retained layer was converted to z-scores (mean 0, SD 1) using the sample statistics, so layers with different units contribute equally to the model.
- **Outputs:** a 19-band GeoTIFF, a channel-order file, the z-score statistics, a mask raster (1 = inside MLRA 42B, 0 = buffer), and diagnostic files (VIF results, correlation matrix, missing-data report).

**Table 3.** Layers removed by the VIF screen.

| Layer | VIF | Interpretation |
|---|---|---|
| prism_tmean (mean temperature) | 268,093 | PRISM computes mean temperature from maximum and minimum temperature, so it is an exact duplicate of information already present. |
| meanelev_32 (mean elevation) | 81.3 | Almost fully predictable (~99%) from the other terrain and climate layers. |
| claytotal_15_cm_p (clay at 15 cm) | 12.4 | Nearly identical to clay at 30 cm; adjacent SOLUS depths are highly correlated. |
| prism_tmin (minimum temperature) | 6.2 | Largely predictable from maximum temperature and elevation-related layers. |

## Solution: Preliminary Model

The preliminary solution is a convolutional autoencoder trained on patches of the covariate stack, implemented in Python with PyTorch (03_autoencoder_first_test.py). An autoencoder learns without

labels: it compresses each patch into a short numeric summary (a latent vector, or "fingerprint") and is trained to rebuild the original patch from that summary. The loss is the difference between the rebuilt and original patch, so the input serves as its own target. Because the fingerprint is far smaller than the patch, the model must learn the dominant spatial and environmental patterns. Fingerprints are then clustered into candidate ecological site groups.

- **Input patches:** 32 × 32 pixel patches (~960 m across) of all 19 channels, centered inside MLRA 42B. Patches containing missing cells are rejected. The first test uses 10,000 patches.
- **Spatial validation split:** patches are assigned to training or validation by 128 × 128-pixel (~3.8 km) spatial blocks, with 20% of blocks held out, so validation patches are not near-copies of training patches.
- **Encoder (CNN):** three blocks of 3 × 3 convolution, batch normalization, ReLU activation, and 2 × 2 max pooling (32, 64, and 128 filters), reducing each patch from 32 × 32 to 4 × 4 pixels, followed by a dense layer producing a 16-number fingerprint.
- **Decoder (CNN):** mirrors the encoder with a dense expansion layer and three upsampling-plus-convolution blocks, ending in a 3 × 3 convolution that outputs all 19 channels.
- **Training:** mean squared error reconstruction loss, Adam optimizer (learning rate 0.001; Kingma and Ba, 2015), batch size 128, learning rate halved when validation loss plateaus, and early stopping after 8 epochs without improvement. Flip and rotation augmentation is not used because aspect-derived channels change meaning when a patch is rotated.
- **Diagnostics:** training and validation loss curves, reconstruction $R^2$ for each channel (to show whether any layer is being ignored), and example original-versus-rebuilt patches.
- **Clustering:** k-means on the fingerprints for k = 3–12 in the first test (to be extended to k = 20), recording silhouette scores and cluster sizes. The number of clusters will be chosen using silhouette, elbow, and stability criteria fixed before any comparison with mapped ecological sites, so that the ecological site map remains an independent check.
- **Baseline:** PCA followed by k-means on the center-pixel values of the same patches (no spatial context). Agreement between autoencoder and baseline groupings is measured with the adjusted Rand index; if the autoencoder groups are no more ecologically coherent than the baseline, the spatial model adds little.
- **Map preview:** patch centers with their cluster labels are exported as a point file for inspection in GIS. A full wall-to-wall map is the next step.

## Code and Reproducibility

All code is in the repository's src folder and runs in order: 01_clip_covariates.R (clipping) and 02_build_covariate_stack.R (stack building. Each script has a clearly marked settings section for file paths and options. The raster data are too large for the repository; the README lists the data sources (PRISM, SOLUS, STEDIS, NRCS MLRA boundaries) and how to rebuild the stack. Small diagnostic outputs (manifest, VIF results, correlation matrix, z-score statistics, channel order) are included.

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[ADD: citation for the STEDIS terrain and spectral derivatives from their documentation.]